Zoo DEX — Native Securities Trading on Lux Quasar with Liquidity Protocol Integration

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Antje Worring & Zach Kelling

Zoo Labs Foundation

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Abstract

The Zoo DEX launches on April 20, 2026 with native trading of equities, fixed income, real-world assets, and tokenised derivatives, all settling on Zoo D-Chain (a Lux D-Chain white-label per LP-134) with sub-second Quasar 4.0 triple-cert finality. The DEX integrates the Liquidity Protocol formally launched by Liquidity.io on April 1, 2026 and adopted by Lux, Hanzo, Zoo, and Pars on April 20, 2026 [11], providing compliance-by-construction through a dedicated Liquidity precompile that applies KYC/AML at the perimeter (SEC ATS, FINRA broker-dealer, and registered transfer agent obligations are met by the Liquidity.io control plane). The architecture is built on Zoo 4.0's GPU-native sovereign L1 [15], activated February 14, 2026 and inheriting the full post-quantum stack from Zoo 3.0 [14]. The DEX ships AMM V2 (constant product) and V3 (concentrated liquidity), with V4 implemented as a precompile to the Lux DEX cross-chain liquidity router; Zoo additionally exposes an optional in-chain local DEX precompile for Zoo-specific assets (per-LLM tokens, fractional NFTs). Order types include limit, market, stop, IOC, FOK, post-only, hidden, and iceberg, with both batch-auction and continuous-limit-book matching. A Robinhood-style retail UX delivers zero commission, fractional shares, 24/7 markets, and sub-1.1s settlement. Optional FHE-encrypted orderflow runs through F-Chain (LP-013); ZK selective disclosure runs through Z-Chain (LP-063). Quantum-secure trading is end-to-end: every trade settles under a triple-cert (BLS$_{12\text{-}381}$ + Ringtail + ML-DSA). The DEX is the migration destination for NFT markets and the realisation of the original "NFT Liquidity Protocol" coinage from the October 31, 2021 Zoo whitepaper [12].

Keywords: Zoo DEX, native securities, Liquidity Protocol, Quasar 4.0, real-world assets, tokenised derivatives, post-quantum trading, Robinhood-style retail, FHE orderflow

Contents

1 Architecture

The Zoo dex runs on Zoo D-Chain — a white-label resell of the Lux D-Chain template defined by LP-134 [8] — which sits inside the Lux chain topology alongside P-, C-, X-, Q-, Z-, A-, B-, M-, and F-Chains. The white-label model gives Zoo chain-local control of asset listings, fee policy, market hours, and validator set while inheriting the Lux consensus, bridge, privacy, and AI substrate without re-implementation.

1.1 Why D-Chain (white-label)

A native trading venue for tokenised securities needs three properties simultaneously:

1. **Sub-second finality** so that retail order flow does not wait for cross-chain confirmations. Quasar 4.0 triple-cert finality on Lux [3] delivers < 1.1s.
2. **Compliance perimeter** so that the chain can enforce regulatory constraints (KYC, AML, accredited-investor checks) without leaking PII into block contents. The Liquidity precompile (§3) provides this.
3. **Sovereign listing policy** so that Zoo can list Zoo-native securities (per-llm tokens, conservation bonds, fractional nfts) without coordinating with the broad Lux validator set on every market opening.

Lux D-Chain template provides the first two; the white-label resell adds the third by giving Zoo a sovereign chain instance with its own governance.

1.2 Position in the Zoo stack

Layer	Role
Zoo 4.0 L1 [15]	General-purpose gpu-native sovereign L1; `$ZOO` gas; per-llm chain hub.
Zoo D-Chain	Specialised D-Chain instance for tokenised securities. White-label of Lux D-Chain template.
Zoo dex	Application contracts on Zoo D-Chain (amm V2, V3, V4, order book, order types).
Liquidity precompile	Compliance perimeter; KYC/AML; broker-dealer hooks; transfer-agent hooks.

1.3 AMM V2 / V3 / V4

V2 — constant product ($x \cdot y = k$). Pools are pairs of tokenised assets (security pair, security/stablecoin, security/$ZOO). LP shares mint as receipts; default fee 30 bps with per-pool override governance. Uniswap-shaped surface so existing analytics, audits, and tooling translate.

V3 — concentrated liquidity. Concentrated-liquidity tick-based pools per the Uniswap v3 design [19]. Multiple fee tiers (5/30/100 bps) coexist for the same pair. Tick crossings, fee accumulation, and range-position accounting run as gpu kernels per LP-009 [1] and LP-132 [6].

V4 — precompile to Lux dex. V4 is not a separate AMM. V4 is the precompile interface that lets Zoo D-Chain contracts route swaps to the Lux dex [18] via B-Chain bridge messages. Use V4 when the deepest liquidity for a pair lives on Lux (e.g., cross-ecosystem assets, institutional CLOB flow, cross-chain routing).

Optional local Zoo dex precompile. For Zoo-internal use cases that want CLOB semantics without the B-Chain hop, Zoo ships an in-chain dex precompile that implements the same matching primitives as the Lux dex but against Zoo D-Chain order books. The precompile shares its kernel implementation with Lux dex; only the venue and gas accounting differ.

1.4 Determinism

The dex runs deterministically on the gpu per LP-132 [6] and the gpu-Residency Invariant of LP-137 [9]: order matching, fee calculation, and tick crossing all stay on-device for the duration of a block. The block-level matching engine reads the order book, runs the matching algorithm (price-time priority for continuous limit book; uniform-price discovery for batch auctions), updates pool state, and emits the trade events all in device memory; only the final cert and event log leave the device.

1.5 Settlement

A trade is final when:

1. Zoo D-Chain produces a block containing the matched trades.
2. Lux Quasar 4.0 emits a triple-cert (bls + Ringtail + ml-dsa) over the D-Chain block hash [3].
3. The trade event is recorded in the chain log; transfer-agent hooks fire (§3); the beneficial owner record is updated.

End-to-end latency target is below 1.1 seconds from order submission to triple-cert finality. The latency is dominated by the consensus round; the matching engine itself completes in microseconds for typical block contents.

1.6 Cross-chain plumbing

- **B-Chain (lp-134).** Cross-chain trade routing, bridge messages between Zoo D-Chain and Lux C-Chain or external chains.
- **F-Chain (lp-013).** Confidential orderflow: ciphertext orders, ciphertext fills, encrypted balances per Confidential ERC-20.
- **Z-Chain (lp-063).** zkp verification for accreditation proofs, geographic eligibility, source-of-funds attestations.
- **A-Chain (lp-134).** TEE-attested AI mining and inference receipts that pay validators in `$AI` alongside the trading-fee share denominated in `$ZOO`.
- **M-Chain (lp-134).** Validator-set custody and key rotation under Lux threshold ceremony rules (LP-017).

1.7 Why this is the right architecture

A pure-app DEX on ZOO 4.0 L1 would work for ZOO-native pairs but would not give the chain a clean compliance boundary against external regulators. A pure routing layer to LUX DEX would be fast but would not let ZOO list its own securities sovereignly. A bespoke chain built from scratch would duplicate two years of LUX consensus, bridge, and privacy work. The white-label D-Chain is the right shape: sovereign listing and policy on top of a battle-tested LUX substrate.

2 Native Securities

The ZOO DEX lists tokenised securities as first-class assets, not as wrapped derivatives or synthetic exposures. Each security class settles directly on ZOO D-Chain with QUASAR 4.0 triple-cert finality [3].

2.1 Equities (fractional shares)

Tokenised equity positions, fractional down to one millionth of a share, settle on ZOO D-Chain. The transfer-agent hook in the LIQUIDITY precompile (§3) updates the official record at LIQUIDITY.io's registered transfer agent on every trade. A holder's beneficial-owner record is updated within the same block as the trade. Voting rights on equity tokens propagate to the holder's wallet via the same record.

Fractional liquidity. A position of 0.0001 shares is tradeable. The matching engine has no minimum-quantity floor; the only floor is the gas cost per trade, which under ZOO 4.0 GPU-native gas is sub-cent for typical order sizes.

Corporate actions. Splits, dividends, mergers, and tender offers propagate via on-chain events emitted by the issuer's LIQUIDITY-attested action contract. Holder wallets reconcile automatically.

2.2 Fixed income

Corporate bonds. Tokenised corporate debt with on-chain coupon distributions. Issuer deposits coupon principal in a custody contract; the contract distributes pro-rata to holders on the coupon date; holders' wallets reflect the distribution as native `$ZOO`-denominated income (or a configurable denomination per bond series).

Treasuries. Tokenised US Treasury exposures (T-bills, notes, bonds) settled through the LIQUIDITY broker-dealer interface. Real-time mark-to-market against published rates.

Yield strips. Coupon-stripping into separately tradeable principal and interest tokens. Useful for laddered yield products served to retail.

2.3 Real-world assets (RWA)

REITs. Tokenised real estate trust shares with quarterly distributions.

Commodities. Tokenised gold, silver, copper, energy — spot exposure backed by warehouse-attested holdings. The attestations live on A-Chain [8] so verification of physical backing is on-chain.

Private equity. Tokenised LP interests in private equity funds, gated by accredited-investor status checks via the Z-Chain selective disclosure path (§7).

Carbon and conservation credits. On-chain carbon credits and conservation impact tokens (consistent with Zoo Labs Foundation's 501(c)(3) mission), tradeable like any other asset and burnable on-chain when retired against an emissions or impact reporting obligation.

2.4 Tokenised derivatives

Perpetual futures. Continuous funding-rate perpetuals on equity, commodity, and crypto underlyings. Funding-rate calculation, mark-price oracle, and liquidation engine all run on-chain. Cross-margined accounts permit capital efficiency across positions.

Dated futures. Calendar futures on commodities and equity indexes, with monthly expiry rolls and physical or cash settlement at expiry.

Options. European-style options on supported underlyings with on-chain exercise. American-style and exotic structures gated by regulator approval and Liquidity broker-dealer review.

2.5 Compliance-by-construction

Every security listing requires:

1. Issuer onboarding through Liquidity.io (the SEC ATS, FINRA broker-dealer, and registered transfer agent control plane).
2. A `SecurityClass` contract on Zoo D-Chain that delegates the regulator-relevant checks (transfer eligibility, holder eligibility, jurisdiction) to the Liquidity precompile.
3. A continuous attestation feed from the issuer's transfer agent to A-Chain, which the dex reads before allowing listing changes.
4. Z-Chain zkp circuits for any holder-side eligibility checks where the holder must prove a property (accreditation status, geography, source-of-funds).

Trades on a security class that has lost good-standing status (e.g., a delisting, a corporate-action freeze, a regulatory hold) are automatically blocked at the precompile level; the matching engine does not even see the order. This is the compliance-by-construction discipline: the rules are not policy in a contract; they are gates in the precompile that the matching engine cannot bypass.

2.6 Settlement economics

T+0 settlement is the default. The Quasar 4.0 cert provides sub-1.1s finality [3]; the transfer-agent hook fires within the same block; the official record is updated atomically with the chain state. There is no T+1 / T+2 settlement window because there is nothing to wait for.

2.7 Why this is materially different from a CEX listing

A CEX listing tokenises a hold-on-balance-sheet position. The exchange owns the asset; the user owns a claim against the exchange. A Zoo D-Chain listing is the asset. The transfer-agent record *is* the chain record. There is no exchange-balance-sheet intermediary. This is what "native securities" means in this paper: the chain is the ledger of record, not a mirror of one.

3 Liquidity Protocol Integration

The Liquidity Protocol is the compliance-by-construction settlement standard for tokenised securities formally launched by Liquidity.io on April 1, 2026 with a soundness proof published alongside [10]. Lux, Hanzo, Zoo, and Pars adopted the Liquidity Protocol on April 20, 2026; this section documents Zoo's adoption [11].

3.1 What the Liquidity Protocol provides

Liquidity Protocol is the contract a chain accepts to integrate with the Liquidity.io control plane. The contract specifies:

1. The KYC/AML perimeter: where checks happen, what they verify, what evidence is admissible.
2. The broker-dealer interface: which actions require BD signoff, what the BD's signing key looks like on-chain, how BD attestations flow through transactions.
3. The transfer-agent interface: how the official record is updated, what events constitute a record update, how the chain's view of beneficial ownership is reconciled with the TA's view.
4. The market-center interface: what data the venue must publish (best-bid-best-offer, last-trade, volume), what venue-side reporting goes back to the BD and TA.
5. The dispute interface: how a flagged transaction is unwound, what cancel/reverse semantics look like at the chain level.

The soundness proof in [10] establishes that an integrating chain that follows the contract cannot construct a state that is unattributable, unrecoverable, or non-compliant under SEC ATS, FINRA BD, and registered TA rules. Zoo inherits this soundness by integrating against the formal contract.

3.2 The Liquidity precompile on Zoo D-Chain

Zoo D-Chain ships a precompile at a reserved address that implements the chain-side half of the Liquidity Protocol. The precompile interface exposes:

```
namespace zoo::dchain::liquidity {

  // KYC/AML perimeter
  bool is_kyc_clear(Address holder);
  bool is_aml_clear(Address holder, Hash transaction_intent);

  // Broker-dealer attestations
  bool bd_authorised(Address holder, ActionId action);
  Hash bd_signoff_required_for(ActionId action);

```

```
  // Transfer agent
  void notify_ta_of_trade(TradeEvent ev);
  Hash ta_record_root_at_block(BlockHeight h);

  // Market center
  void publish_bbo(Pair p, Price bid, Price ask);
  void publish_last_trade(Pair p, Price px, Quantity qty);

  // Dispute / unwind
  bool is_action_disputed(Hash action_hash);
  void execute_unwind(Hash action_hash, UnwindReason reason);
}
```

Listing 1: Liquidity precompile (Zoo D-Chain)

The precompile is the only on-chain entity that can mutate the `SecurityClass` state for a regulated security. The matching engine, the AMM, the order types — all of them call into the precompile to verify each action before applying state changes. Trade requests for which the precompile returns "not allowed" never reach the order book.

3.3 Cross-chain trade routing through B-Chain

For pairs where the deepest liquidity lives outside Zoo D-Chain (e.g., a security listed on Zoo but with cross-listing liquidity on Lux C-Chain or external venues), trades route via B-Chain [8]:

1. Order submitted on Zoo D-Chain.
2. Matching engine determines best route includes a cross-chain leg.
3. B-Chain bridge message emitted; Lux dex [18] executes the cross-chain leg.
4. Settlement leg returns via B-Chain; the cross-chain attestation carries triple-cert proof.
5. Local fill is recorded; transfer-agent notify fires; trade finalised.

The Liquidity precompile is consulted on both sides of the B-Chain hop: Zoo D-Chain enforces Zoo-side compliance; Lux C-Chain (also a Liquidity Protocol integrator) enforces Lux-side compliance. A trade that violates compliance on either side is rejected.

3.4 Why a precompile and not a smart contract

The Liquidity integration must be unbypassable. If KYC checks lived in an EVM contract, an application contract could fork the precompile, omit the KYC call, and trade against the matching engine directly. By living in the precompile (chain-built-in code, not EVM-deployable), the Liquidity integration is enforced at the chain interpreter level. There is no application-contract path that the matching engine accepts that does not first go through the precompile.

This is the compliance-by-construction property that the Liquidity Protocol soundness proof relies on.

3.5 Soundness inheritance

The Liquidity Protocol formal proof [10] shows that a chain that:

1. exposes the Liquidity precompile as the only mutator of `SecurityClass` state, and
2. routes every trade, transfer, and corporate action through the precompile, and
3. maintains the cross-chain invariants for B-Chain hops,

inherits the soundness property: no chain state is reachable that violates the SEC ATS, FINRA BD, or registered TA obligations encoded in the protocol. Zoo D-Chain meets all three preconditions and therefore inherits soundness from April 20, 2026 onward.

3.6 Operational interfaces

- **Issuer onboarding.** Issuer registers with Liquidity.io; the resulting attestation is published on A-Chain; Zoo D-Chain's listing-admission flow reads the A-Chain attestation; once verified, the listing becomes active.
- **Holder onboarding.** Holder onboards via wallet UX through Liquidity.io's KYC flow; the resulting on-chain record sets a holder-eligibility bit on A-Chain that the precompile reads.
- **Audit interface.** Regulators query the on-chain record directly; the chain log is the audit trail; no secondary database is needed.

4 Order Types and Matching

Zoo dex ships the full retail and institutional order-type matrix on day one of the April 20, 2026 launch.

4.1 Order types (taker side)

Limit. Buy or sell at a specified price or better. Resting limit orders post to the continuous limit book at the specified price level and match by price-time priority.

Market. Buy or sell at the best available price. Routes against the AMM pool when the AMM has a better price than the top of the book; uses the order book otherwise.

Stop. Triggered when the market reaches the stop price; converts into a market or limit order at the trigger.

IOC (Immediate-or-Cancel). Match what can be matched at the specified price; cancel any unfilled remainder. No resting state.

FOK (Fill-or-Kill). Match the full quantity at the specified price or cancel the entire order. No partial fills.

Post-only. Add liquidity to the order book at the specified price; reject if the order would cross the spread (i.e., act as a taker). Useful for makers earning rebates.

Hidden. Resting limit order that does not display in the public order book. Visible to the matching engine only. Used by institutional flow that does not want to telegraph size.

Iceberg. A large limit order that displays only a fraction of its size in the public book. As the displayed slice fills, the next slice becomes visible. Useful for working size without spooking the book.

4.2 Continuous limit book

The continuous limit book is the high-frequency matching engine. It implements price-time priority: among orders at the same price, the earliest-arrived order matches first. The order book lives in device memory per LP-137 [9]; updates run as GPU kernels.

A typical continuous-book trade lifecycle:

1. Order arrives at the sequencer; passes the LIQUIDITY precompile check.
2. Order is inserted into the per-pair price-time-priority order book.
3. If the order crosses an existing book level, the crossing match is generated; both orders' resting states are updated; the trade event is emitted.
4. Trade event flows through the transfer-agent notify path [10].
5. Block production seals the trades into a QUASAR 4.0 triple-cert.

4.3 Batch auctions

For fair price discovery on instruments where front-running risk is material (illiquid securities, opening prints, closing prints), the DEX also runs batch auctions. A batch auction:

1. Accepts orders during the batch window (e.g., 100ms).
2. At the window close, computes the uniform clearing price that maximises matched volume.
3. Executes all crossing orders at the uniform clearing price.
4. Cancels any orders that do not match at the clearing price (or rests them in the continuous book if so flagged).

Batch auctions eliminate the latency arms race because the inside order does not benefit from being earlier within the batch; only prices and quantities matter, not microsecond arrival times. This is materially better for retail than continuous matching for the same instrument set.

4.4 When to use which

Mode	**Use when**
Continuous limit book	Liquid pairs, high-frequency strategies, market-makers, retail at-the-touch trading.
Batch auction	Opening prints, closing prints, illiquid securities, retail order flow that benefits from front-running protection.

AMM (V2/V3)	Long-tail pairs, programmatic-LP yield, automated market-making without an active maker.

The matching engine routes a market order across all three venues and executes against whichever has the best price; the user receives a single fill report.

4.5 Self-trade prevention

Two orders from the same beneficial owner cannot match each other. The matching engine consults the LIQUIDITY precompile's beneficial-owner record; if a match would put both sides on the same beneficial owner, the engine cancels the resting side and applies the inbound side as a new resting order. This prevents wash trading without requiring the user to explicitly tag orders.

4.6 Order routing across V2/V3/V4

A retail market order for a security pair flows through the router:

1. Router queries V2 pool, V3 pool, V4 (Lux DEX), and the local CLOB precompile for executable price at the order size.
2. Router computes the price-impact-adjusted best venue (or splits across venues for large orders).
3. Router executes against the chosen venue(s); each leg goes through the LIQUIDITY precompile check.
4. Router returns a single weighted-average fill report to the user.

This is invisible to the user; it is a router-side concern.

4.7 Cancel-replace semantics

A cancel-replace transaction atomically cancels the existing order and submits a new one. If the cancel succeeds and the replace fails (e.g., the precompile rejects the new order), the engine reverts to a clean state with no resting order. There is no half-applied state.

4.8 Reporting

Every executed trade emits a `TradeEvent` on chain that includes:

- Pair, price, quantity, side, taker/maker flag.
- Beneficial-owner reference (or its hash, in privacy mode).
- Venue (V2 / V3 / V4 / CLOB).
- LIQUIDITY-precompile attestation hash.
- QUASARCert reference (for finality verification).

Public market-data subscribers (the BBO, the last-trade tape, the volume by venue) consume these events. There is no separate market-data infrastructure; the chain log is the tape.

5 Robinhood-Style Retail UX

The Zoo DEX is engineered to be the simplest, fastest, and cheapest place for a retail trader to access tokenised global markets. This section documents the user-facing properties.

5.1 Zero commission

Trades on Zoo DEX cost zero commission for retail users. The fee schedule:

- **Maker rebate.** Resting limit orders that add liquidity to the book earn a small rebate funded by the taker fee.
- **Taker fee.** Marketable orders pay a small fee (default 5 bps for retail, configurable per security class by Zoo DAO governance).
- **No commission.** The retail user does not pay a per-order commission on top of the spread + taker fee. The spread itself is the entry/exit cost.
- **No PFOF (payment for order flow).** The chain is the venue; there is no off-chain market-maker paying for the right to internalise the order.

The taker fee is split between the protocol treasury (for ongoing protocol development), validator rewards (block-production incentive), and a conservation allocation (consistent with Zoo Labs Foundation's 501(c)(3) mission).

5.2 Fractional shares

Every listing is fractionable down to one millionth of a share. A retail user with $10 to invest can buy fractional positions in any listed equity, ETF, or REIT. There is no minimum order size other than the per-trade gas floor, which under Zoo 4.0 GPU-native gas is sub-cent for typical trade sizes.

Dividend and corporate-action distributions on fractional positions are computed pro-rata to the holder's exact fractional balance and distributed automatically by the issuer's action contract.

5.3 24/7 markets

The chain runs continuously. Securities trade outside traditional market hours for instruments whose issuer or transfer agent has opted into 24/7 trading. For instruments that have not opted in (e.g., where the underlying market hours are constrained by a regulator obligation), the chain enforces market hours at the Liquidity precompile: trades outside the listing's hours window are rejected.

The matching engine does not pause; only the per-listing eligibility bit changes. This means a 24/7 instrument continues to match while a market-hours-constrained instrument is paused, all in the same block.

5.4 Instant settlement

Sub-1.1s end-to-end latency from order submission to triple-cert finality [3]. Practically:

- User taps "buy".
- Wallet signs and submits to D-Chain sequencer.

- Matching engine fills within microseconds.
- Quasar 4.0 cert seals within sub-second window.
- Wallet shows "filled, final" to the user.

There is no T+1 / T+2 settlement clearance. The official record at the transfer agent is updated atomically with the trade. The user's beneficial ownership of the new position is final at block close.

5.5 Wallet UX

The wallet is the user's interface to the DEX. The relevant panes:

- **Portfolio.** List of positions with current price, unrealised P&L, today's change.
- **Trade.** Symbol search, order entry, order book or AMM quote view, order types selector.
- **History.** Filled and unfilled orders with on-chain attestation links.
- **Tax.** On-chain cost-basis tracking; per-jurisdiction tax-event reports generated from the chain log.
- **Privacy.** Toggle for confidential balances (F-Chain FHE) and selective-disclosure proofs (Z-Chain ZKP).

The wallet does not run separate market-data infrastructure; it subscribes to the chain log directly via standard Lux Wallet RPC (per LP-105 [5]).

5.6 Mobile parity

The wallet ships in three form factors — browser extension, mobile (iOS/Android), and Lux Desktop — with the same screen inventory and identical signing semantics. A trade started on mobile and finished on desktop signs identically; the chain treats the order regardless of the form factor of the signer.

5.7 Make money like the 1%

The pitch to retail: *the same instruments and the same markets that institutional capital trades, on the same venue, with the same latency, at no commission.* Tokenised treasuries pay the same yield to a $50 holder as to a $50M holder. Fractional REITs distribute pro-rata. Tokenised perpetuals fund identically per side. The barrier to access has historically been intermediary fee stacks and minimum-account-balance rules. Zoo DEX's zero-commission, fractional, 24/7, on-chain T+0 settlement removes those barriers entirely.

5.8 No dark patterns

- No PFOF: the chain is the venue; nobody pays to internalise retail flow.
- No spread mark-ups: the spread the user sees is the spread the matching engine sees.
- No order-routing rebates that misalign with user execution quality: the router optimises for user fill price, period.

- No gamification of trading frequency: the wallet does not push trade volume; defaults are set toward longer-horizon positions.

This is design-time alignment with consumer protection, not just post-hoc compliance.

5.9 Education and risk disclosures

Every onboarding flow includes risk disclosures appropriate to the asset class. Derivatives onboarding requires explicit acknowledgement of leverage and liquidation mechanics. Accreditation-gated instruments require Z-Chain ZKP proof of accredited status before the wallet exposes the instrument in search.

6 Quantum-Secure Trading

Every trade on Zoo DEX settles under a Quasar 4.0 triple-cert inherited from the Lux L1 [3]. The cert composes three independent signature layers; a trade is final only when all three have signed.

6.1 The triple cert at the trade level

A finalised trade carries:

1. **BLS$_{12\text{-}381}$ aggregate.** 48 bytes. Classical fast-path proof from the Lux validator quorum. Useful for the low-latency check that gates pre-final block propagation.
2. **Ringtail threshold.** Post-quantum threshold proof, Ring-LWE-based. ∼2.0–2.4 KB.
3. **ML-DSA-65 identity proof.** FIPS 204 conforming, ∼3.3 KB per validator. Establishes which validator authorised the cert layer.

A trade with only BLS is not final under Zoo DEX; the matching engine waits for the full triple-cert before marking the trade as settled.

6.2 Q-day resilience

Under a near-term Q-day adversary — a cryptographically relevant quantum computer (CRQC) capable of breaking ECDSA and pairing-based schemes — the trade record continues to validate because the Ringtail and ML-DSA proofs do not depend on classical hardness assumptions. A trade settled on Zoo DEX on April 20, 2026 remains independently verifiable post-Q-day; the chain does not need a flag day to invalidate pre-Q-day records.

This matters for tokenised securities specifically because beneficial-ownership records persist for the lifetime of the asset (potentially decades for fixed-income or equity holdings). A classical-only signature on a 2026 trade in a 30-year corporate bond would be retroactively forgeable by a CRQC available in, say, 2034. The triple-cert design eliminates that exposure.

6.3 Harvest-now-decrypt-later resistance

A passive observer can record every public on-chain transaction today and decrypt the classical proofs later when a CRQC becomes available. This is the harvest-now adversary. The triple-cert discipline means the harvest-now record is not exploitable: even once BLS falls, the Ringtail and ML-DSA proofs continue to authenticate the trade. The settlement record is immune to the harvest-now class.

6.4 Validator key management

Validators run dual-key wallets:

- Classical key for BLS operations (fast path).
- Threshold ML-DSA-65 key for identity proofs.
- Threshold Ringtail key for post-quantum threshold operations.

Key rotation runs on the LP-017 schedule: every 90 days the validator set runs a re-keying ceremony on M-Chain [8]. Stake is anchored to the *validator*, not the per-epoch key, so rotation is non-disruptive to staked-balance accounting.

6.5 Trade-finality semantics

A wallet showing "filled, final" to the user is showing the following on-chain state:

1. The trade event is in a block sealed by Zoo D-Chain.
2. The block hash is referenced inside a QuasarCert.
3. The QuasarCert carries valid signatures on all three lanes (BLS, Ringtail, ML-DSA).
4. The transfer-agent notify has fired and been acknowledged.

If any of these four conditions fail at the time of display, the wallet shows "filled, pending finality". Time-to-final is sub-1.1s in normal operation.

6.6 Light-client verification

A retail wallet does not have to trust a full node to verify finality. The triple-cert verifier is small enough to run inside the wallet:

```
bool verify_trade_finality(
    TradeEvent ev,
    QuasarCert4 cert,
    ValidatorSet vs
) {
    // Block contains the trade
    if (!merkle_proof_valid(ev, cert.block_root)) return false;
    // Triple-cert valid against the validator set
    if (!bls_verify(cert.bls, cert.block_root, vs))           return false;
    if (!ringtail_verify(cert.ringtail, cert.block_root, vs)) return false;
    if (!mldsa_verify(cert.mldsa, cert.block_root, vs))       return false;
    // Liquidity precompile attested the trade
    if (!liquidity_attestation_valid(ev.attestation)) return false;
    return true;
}
```

Listing 2: Light-client cert verifier

Practical wallet verification budget is well under 100ms per trade on consumer hardware (the dominant cost is the ML-DSA aggregate verification scaling with the threshold size).

6.7 Disaster recovery

- **Single lattice scheme break.** If a published break of either Ringtail or ML-DSA surfaces, trading halts pending parameter re-issuance. Existing settled trades remain valid because they carry the unbroken lane plus the classical lane; new trades require new parameters.

- **Both lattice schemes break.** Trading halts; the DAO convenes an emergency response. Pre-existing trades retain partial validation (classical lane), but new trades are paused until a new PQ scheme is integrated.

- **Classical scheme break only.** Trading continues; the classical lane is downgraded to advisory; PQ lanes continue to authenticate. This is the steady-state under Q-day.

6.8 Why this matters for the asset classes the DEX lists

Equity, fixed income, RWA, and derivatives all carry long-lived ownership records. A retail user buying tokenised treasuries with a 10-year term needs to know the record of that purchase will be verifiable for the term of the bond. The triple-cert is the minimum bar for that guarantee.

The Zoo DEX's value proposition — "make money like the 1% finally" (§5) — is meaningless without end-to-end Q-day-resilient settlement. A retail user holding a 30-year corporate bond as a tokenised position must be confident that the position record is valid when the bond matures, regardless of cryptographic developments in the intervening decades. The triple-cert is what makes that promise creditable.

7 Privacy

Zoo DEX ships two complementary privacy paths via the Lux chain topology: optional FHE-encrypted orderflow via F-Chain [2], and ZKP-based selective disclosure via Z-Chain [4].

7.1 F-Chain FHE for orderflow

F-Chain (FVM) is a fully-homomorphic-encryption execution environment [2]. Zoo DEX contracts can dispatch orderflow to F-Chain so that orders, balances, and fills are ciphertexts. A confidential trade looks like:

1. User's wallet encrypts the order under the F-Chain public key. The cleartext (price, quantity, side) never leaves the wallet.

2. The F-Chain contract receives the ciphertext order and runs the matching engine over the ciphertexts: comparison, crossing detection, and fill computation are all ciphertext operations.

3. The fill receipt is returned to the user as a ciphertext; the user decrypts locally to see the fill price and quantity.

4. The trade is anchored to D-Chain with a public commitment (the encrypted balance update), but no observer can read the cleartext order or fill.

Use F-Chain confidential orderflow when:

- Front-running risk is material (illiquid securities, institutional block flow, accumulation strategies).

- Privacy of the position itself is part of the user's compliance posture (treasury holdings disclosed only at regulator-defined intervals).
- Per-LLM token holdings should not telegraph governance positioning.

Confidential balances (LP-067). The position itself can live as a Confidential ERC-20 balance. Holders' balances are ciphertexts; transfers update ciphertexts without revealing amounts. The Liquidity precompile can verify holder eligibility on the cleartext side via TEE-attested decryption where regulator obligations require it (e.g., transfer-agent updates need cleartext beneficial-owner identification, but the publicly visible chain log only carries ciphertexts).

7.2 Z-Chain ZKP for selective disclosure

Z-Chain (ZVM) is a zero-knowledge proof execution and registry environment [4]. Zoo dex dispatches verification jobs to Z-Chain via precompile.

Accreditation proofs. A user proves "I am an accredited investor" without revealing income, net worth, or jurisdiction. The proof is generated against a circuit that takes the user's Liquidity-issued accreditation attestation as a private witness and outputs a public "yes".

Geographic eligibility. A user proves "I am not in a sanctioned jurisdiction" without revealing the actual jurisdiction. The proof is generated against a circuit that takes a geographic-attestation private witness and outputs the negative-list result.

Source-of-funds. A user proves "these funds came from a regulated source" without revealing the source identity or transaction history. Used for AML compliance on large positions.

Rolled-up KYC. For high-volume venues, Z-Chain rolls up batches of KYC eligibility checks into a single Groth16 verification. The matching engine consults the rolled-up proof rather than per-order proof verification.

7.3 When to use which

Property to protect	F-Chain (FHE)	Z-Chain (ZKP)
Hide order price/qty	Yes	No
Hide balance	Yes (LP-067)	No (only the witness)
Prove accreditation	Indirect (via TEE)	Direct (predicate)
Hide identity	Via key management	Yes (predicate over identity)
Per-op cost	High	Low to moderate
Best for	Confidential orderflow, balances, block flow	Eligibility predicates, regulatory checks

The wallet exposes a privacy-mode toggle (§5) that picks the right chain for the user's intent. "Hide this trade" routes to F-Chain confidential orderflow. "Prove eligibility" routes to Z-Chain selective disclosure.

7.4 Privacy and compliance compose

A common confusion is whether privacy and compliance are at odds. On Zoo DEX they compose:

- The Liquidity precompile sees what it needs to see for regulatory obligations (KYC status, accreditation, transfer eligibility) but does not see public order data.
- Public observers see what they need to see for fair price discovery (BBO, last trade) but do not see the user's identity or holdings.
- Regulators see the full audit trail through the Liquidity Protocol's audit interface, with appropriate authorisation, but do not have a public window into the order book.

The compliance perimeter is at the Liquidity precompile; the privacy perimeter is at the F-Chain / Z-Chain boundary. They are orthogonal axes; both can be enabled simultaneously.

7.5 GPU-native privacy

Both privacy paths are GPU-native end-to-end. F-Chain TFHE bootstrapping runs on the validator GPU per LP-013; Z-Chain Groth16 verification runs on the GPU per LP-063. The privacy primitives in Zoo DEX are not bolted on at application layer; they execute on the same substrate as the rest of the chain.

7.6 Key management for privacy mode

- **F-Chain encryption keys.** Generated client-side; the user holds the decryption key in the wallet. Loss of the wallet means loss of the private balance unless the user has set up social recovery via the Lux Wallet recovery flow.
- **Z-Chain proving keys.** Public-circuit-specific. No user-held secret; the proof generator uses the user's Liquidity-issued attestation as the witness.
- **Wallet privacy mode toggle.** Per-session, with explicit confirmation when toggling to confidential mode for a regulated security (because the wallet warns about the lookup-cost trade-off).

7.7 Audit access

For regulator-authorised audit, the Liquidity Protocol audit interface ([10]) provides a controlled path for cleartext access to the records that matter. The chain log itself does not retain decryption keys; audit access is per-record, authorisation-gated, and logged on chain.

8 Top-2 NFT Migration and the Original NFT Liquidity Protocol

The Zoo DEX is the migration destination for NFT markets, and the realisation of the original "NFT Liquidity Protocol" coinage from the October 31, 2021 Zoo whitepaper [12].

8.1 What the 2021 paper coined

The 2021 Zoo whitepaper introduced "NFT Liquidity Protocol" as the term for a system in which NFTs become productive collateral inside a yield-bearing on-chain market. The 2021 vision had three moving parts:

1. Wildlife conservation NFTs as on-chain collateral.
2. Borrowing and lending against the NFTs without the holder having to sell.
3. Yield mechanisms that fed back to conservation.

The original BSC implementation in 2021 had infrastructural limitations (no native fractional secondary trading, no on-chain appraisal, no privacy primitives, no triple-cert finality). The Zoo DEX on Zoo D-Chain in April 2026 is the first venue where every original 2021 commitment is implementable end-to-end.

8.2 Top-2 NFT migration plan

Zoo 4.0 launched (February 14, 2026) with an explicit go-to-market for NFT migration, targeting the leading NFT collections in wildlife/conservation, AI/agent, and metaverse categories [15]. The Zoo DEX launch on April 20, 2026 makes that go-to-market actionable: the migrating collections now have a venue on which to trade and to borrow against.

8.3 Native fractional NFT trading

A whole NFT on Zoo D-Chain can be wrapped into a Fractional ERC-4626 vault that issues fungible shares. The vault shares trade on Zoo DEX V3 as a concentrated-liquidity pair against $ZOO or stablecoin. Properties:

- **Continuous price discovery.** The vault share's price is a continuous market quote, not an episodic auction.
- **Composable.** The vault shares are ZRC-20 tokens; they compose into AMM pools, lending markets, derivatives, and the Liquidity-precompile-protected transfer flow.
- **Re-aggregable.** Holders accumulating sufficient shares can redeem the underlying NFT from the vault by burning all outstanding shares.

8.4 Collateral-backed lending (the realised 2021 vision)

A holder of a high-value NFT can use it as collateral for a loan in stablecoin, $ZOO, or any listed asset. The flow:

1. Holder locks the NFT in a loan-escrow contract.
2. Loan terms (principal, APR, duration, collateral floor) are agreed with a lender (or pooled-lender market).
3. Counterparties sign the loan agreement under the post-quantum scheme inherited from Zoo 3.0 [14] (Ringtail + ML-DSA-65).
4. Holder receives the loan principal in the chosen denomination.

5. Liquidation oracle (A-Chain attested) monitors the collateral floor against the appraisal feed (which can run confidentially via F-Chain [2]).

6. Holder repays principal plus interest by the maturity date; the NFT is released. Or the collateral floor is breached and liquidation triggers per Zoo 3.0 NFT Liquidity Protocol semantics.

This is exactly the 2021 paper's vision, with the missing infrastructural pieces filled in by Zoo 3.0 (PQ signing) and Zoo 4.0 (GPU-native execution and DEX).

8.5 Provenance for AI/agent NFT collections

Collections whose value derives from on-chain AI provenance (generative-art collections, agent NFTs, on-chain creatures with AI behaviour) benefit from A-Chain-anchored TEE inference receipts [8]. The provenance chain on Zoo D-Chain is verifiable end-to-end: every act of generation that produced a token in the collection is traceable through the inference-receipt chain to a TEE-attested execution.

8.6 Token-bound accounts

ZRC-721 NFTs on Zoo D-Chain have native ERC-6551-style token-bound accounts. The account can hold $ZOO, per-LLM tokens, other NFTs, and accrue yield from staking or LP positions. Combined with the Zoo DEX, the NFT becomes a programmable agent: it can hold a portfolio, execute trades on its own behalf within whitelisted bounds, and participate in governance.

8.7 Migration mechanics

1. **Snapshot.** Zoo Bridge takes a snapshot of holders on the source chain (Ethereum, BSC, etc.).

2. **Lock-and-mint.** The source-chain contract is escrowed via the bridge committee; the Zoo D-Chain native ZRC-721 is minted to the same wallet address.

3. **Metadata mirror.** On-chain metadata moves over with provenance hash chain intact; off-chain assets re-pin to Zoo's decentralised storage layer.

4. **Treasury onboarding.** The collection's treasury is onboarded to a Zoo DAO-governed multisig with optional F-Chain confidential balance.

5. **Royalty redirection.** Future royalties accrue to the D-Chain contract; the bridge keeps the source-chain contract readable for legacy explorers.

The bridge committee under Zoo 3.0 PQ discipline [14] attests the migration; the destination mint carries triple-cert finality.

8.8 Continuous secondary trading

After migration, every transfer of an NFT from the migrated collection trades on Zoo DEX: as a whole-token transfer against a price-discovery pair, as a fractional share via the V3 vault pool, or as collateral inside a loan agreement. The collection's secondary-market liquidity is materially better on Zoo DEX than on its source chain because of:

- Sub-1.1s settlement (vs. multi-block confirmation elsewhere).

- Native fractional secondary market.
- Composable loan agreements.
- Confidential balance and orderflow optionality.

8.9 Why this closes a 2021 commitment

The 2021 Zoo whitepaper coined "NFT Liquidity Protocol" but the BSC infrastructure of the time could not deliver the full promise. Zoo DEX on Zoo D-Chain in April 2026 delivers it: sub-second-final, post-quantum, compliance-perimetered, with native fractional secondary, confidential balances, and a collateralised lending primitive. This paper is the activation document for that closure.

9 Operator Economics

Zoo D-Chain validators stake `$ZOO`; rewards are paid as a share of trading-fee revenue plus `$AI` from compute proofs delegated to the Hanzo AI Chain. This section documents the mechanics.

9.1 Validator stake

Zoo D-Chain validators bond `$ZOO` as their primary stake. The minimum stake is set by the Zoo DAO and revisited on a quarterly governance cadence; at activation it is denominated in a `$ZOO` amount sufficient to align validator incentives with chain integrity but low enough that small-scale validators can participate.

Optional secondary stake of `$AI` is permitted; `$AI` stake gives the validator priority access to AI mining-receipt attribution from the Hanzo AI Chain inference flow that anchors through A-Chain.

9.2 Reward sources

Validators earn from three sources:

1. **Trading-fee share (`$ZOO` or fee-token denominated).** A configurable share of taker fees on every trade flows to validators. Default split at activation: 40% validator pool, 30% protocol treasury, 20% maker rebate, 10% conservation allocation.
2. **Block production rewards (`$ZOO`).** Per-block subsidy that decays on a defined schedule. The subsidy is denominated in `$ZOO` drawn from the treasury per the Zoo DAO schedule.
3. **AI mining receipts (`$AI`).** Validators that operate or delegate to Hanzo AI Chain compute capacity receive a portion of inference receipts as `$AI`. The mining loop binds compute to receipts through TEE attestation (see [15] §AI mining).

9.3 Slashing

Slashing is enforced via Q-Chain ceremonies under LP-105 [5] for double-sign and via A-Chain attestation gaps for liveness failures. Specific slashable events:

- **Double-sign.** Signing two conflicting blocks at the same height. Severe penalty: significant percentage of bonded `$ZOO` burned plus removal from validator set.

- **Liveness failure.** Missing more than the threshold number of blocks within an epoch. Moderate penalty: per-block missed-attestation deduction.
- **Compromised key not rotated.** Failing to rotate a compromised key within the LP-017 schedule [8] after a published advisory. Penalty: removal from validator set.
- **Compliance precompile bypass attempt.** Submitting blocks containing trades that bypass the Liquidity precompile attestation. Severe penalty: full bonded stake burned plus permanent ejection.

The compliance-precompile-bypass slash is the most severe because the soundness of the Liquidity Protocol integration [10] depends on the precompile being the only mutator of `SecurityClass` state. Validators that attempt to bypass it break the soundness contract, so the slash is total.

9.4 Maker incentives

A portion of taker fees flows back to makers as a per-fill rebate. The default maker rebate at activation is 1 bp on 5 bp taker fee (so the maker receives 1 bp per share traded against their resting order; the venue keeps 4 bps for the validator pool, treasury, and conservation allocation). Per-pair rebates can be configured by Zoo dao governance to attract liquidity in specific securities.

9.5 Hanzo AI Chain compute integration

A validator running Zen-family inference workloads inside a TEE-attested container can earn `$AI` simultaneously with running Zoo D-Chain block production. The hardware overlap is deliberate: the QuasarGPU substrate [6] expects validators to run on data-center GPUs with confidential compute; those same GPUs execute Zen inference. A single validator hardware investment serves two revenue streams.

The economic relationship to the Hanzo AI Chain is documented in the Hanzo AI Chain whitepaper; Zoo validators participate through delegation and through co-location of inference workers on the validator hardware.

9.6 Protocol treasury

The protocol treasury is funded from 30% of taker fees by default. It is governed by the Zoo dao and supports:

- Ongoing protocol development (engineering, audits, security reviews).
- Listing-onboarding subsidies for new asset classes.
- Liquidity bootstrapping for newly-listed pairs.
- Insurance fund against validator slashing edge cases that could degrade end-user experience.

9.7 Conservation allocation

Consistent with Zoo Labs Foundation's 501(c)(3) mission, 10% of taker fees flow to a conservation allocation. The allocation is administered by the Zoo Foundation under the same impact-metric oracle and reporting framework as the rest of the foundation's on-chain conservation grants. This is a structural property of the fee schedule, not a discretionary policy: every trade contributes directly to species preservation, with on-chain reporting available for audit.

9.8 Validator hardware budget

- Data-center GPU (H100-class or successor) with confidential compute support.
- NVMe storage for state plus order-book persistence.
- Network: high-bandwidth, low-latency (sub-millisecond to peer set; sub-100ms to global edge).
- Operational redundancy: hot standby, key-rotation tooling, TEE-attestation tooling.

The hardware budget is consistent with Lux validator expectations under Quasar 4.0 (LP-105 [5]), which is intentional: the Zoo D-Chain validator does not have to be a different hardware class from the Lux mainnet validator.

9.9 Economic security at activation

At activation (April 20, 2026), the validator set is sized to provide sufficient stake-weighted security against a single-actor takeover; the threshold for cert finality is $t \geq \lceil 2n/3 \rceil$. The activation validator-set size and per-validator minimum stake are set by the Zoo DAO prior to activation; the parameters are public on-chain.

10 Activation Roadmap

Zoo DEX activates on April 20, 2026, alongside Lux, Hanzo, and Pars adopting the Liquidity Protocol formally launched on April 1, 2026 by Liquidity.io [10, 11].

10.1 Phase 0 — spec freeze (2026-04-01)

This paper, the supporting ZIPs (Zoo DEX V2/V3/V4 contracts, Liquidity precompile interface, retail UX, conservation allocation), and the activation client release freeze on 2026-04-01 (the Liquidity Protocol formal launch). After this date only spec-clarification PRs are accepted; behaviour-changing PRs slip to a post-activation patch release.

10.2 Phase 1 — testnet bring-up (2026-04-01 through 2026-04-13)

- Zoo D-Chain testnet validators come online with activation-candidate binaries.
- AMM V2 / V3 deploy on testnet with seed liquidity for the initial listing set.
- AMM V4 precompile target dummied to a testnet Lux DEX instance for cross-chain test paths.
- Liquidity precompile testnet image deployed; KYC, BD, and TA hooks exercised end-to-end against Liquidity.io sandbox endpoints.
- Initial securities listings onboarded in dry-run mode: equity tokens, treasury tokens, REIT tokens, perpetual contracts on a representative basket.
- Wallet release-candidates exercise the trade flow on testnet for retail UX validation.

10.3 Phase 2 — canary (2026-04-13 through 2026-04-19)

- Zoo D-Chain canary mainnet brought up with a subset of validators.
- Initial market-makers onboarded with rebate-tier agreements.
- Closed-beta retail user cohort tests trade flow on a small subset of listings.
- Cross-chain routing exercised via B-Chain to Lux C-Chain canary.
- Liquidity-precompile attestation flow exercised on production Liquidity.io endpoints.

10.4 Phase 3 — activation (2026-04-20)

- Zoo D-Chain mainnet open to all validators.
- AMM V2 / V3 / V4 production-active.
- Local Zoo dex CLOB precompile activated.
- Continuous limit book and batch auction modes available for retail and institutional flow.
- Order types matrix (limit, market, stop, IOC, FOK, post-only, hidden, iceberg) available across all listings.
- Wallet GA with Zoo dex access.
- Initial securities listing cohort opens for trading: equities (tokenised fractional shares), fixed income (corporate bonds, treasuries), RWA (REITs, commodities, carbon credits), derivatives (perpetuals, dated futures, European options).
- NFT migration second cohort opens for trading on the dex with native fractional V3 pools.
- Conservation allocation begins flowing to the foundation's impact program from the first trade onward.

10.5 Phase 4 — post-activation roll-forward

Zoo dex continues to ship features incrementally. Roadmap items targeted for post-activation patch releases:

- Phased pair listings: weekly cohorts of new tokenised securities listings as issuers complete Liquidity.io onboarding.
- Market-maker programme expansion: tiered rebate schedules, bespoke risk-allocation lines, per-pair maker grants from the protocol treasury.
- Institutional bridges: prime-broker integrations for tokenised-securities flow, custody-vault integrations for large holders, on-ramp/off-ramp via existing exchanges via the Liquidity Protocol's BD interface.
- International listings: cohort-by-cohort onboarding of non-US issuers under the Liquidity Protocol's jurisdiction-aware compliance perimeter.

10.6 Cross-paper relationships

Zoo DEX sits at the intersection of several Zoo papers:

- **Upstream:** Zoo 4.0 [15] (February 14, 2026 L1 graduation). DEX runs on the GPU-native L1 substrate; the AMM curve math, order-book matching, and cert generation all execute on the GPU.
- **Upstream:** Zoo 3.0 [14] (October 31, 2025 full PQ stack). The DEX's settlement Q-day resilience, NFT-loan PQ signing, and validator key management all inherit the 3.0 PQ posture.
- **Lateral:** Liquidity Protocol formal proof [10] (April 1, 2026 Liquidity.io launch). Compliance perimeter and audit interface are inherited.
- **Lateral:** Zoo Adopts the Liquidity Protocol [11] (April 20, 2026 adoption note). The proof artifact for Zoo's specific adoption.
- **Lateral:** Zoo 2025 securities and DAO [17] ($113T market thesis). The market-sizing and governance frame for the DEX's securities listings.
- **Lateral:** Per-LLM chains [16]. Per-LLM token markets list on the DEX alongside traditional securities.
- **Origin:** Zoo 2021 original whitepaper [12]. The DEX is the realisation of the 2021 "NFT Liquidity Protocol" coinage.

10.7 Out-of-scope for this launch

- Synthetic exposures and structured products beyond the listed perpetual / dated future / European option matrix.
- Cross-jurisdictional retail accounts beyond the initial US-centric Liquidity.io perimeter; international retail onboarding ships post-activation.
- Margin lending against tokenised-equity collateral; ships in a follow-up release with dedicated risk parameters.

10.8 One-line summary

Zoo DEX is what happens when you take Zoo 4.0's GPU-native sovereign L1, settle native securities on a Lux-D-Chain white-label, integrate the Liquidity Protocol's compliance perimeter, and ship a Robinhood-style retail UX with zero commission, fractional shares, 24/7 markets, and sub-1.1s T+0 settlement. April 20, 2026.

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